

## Lesson 10 · Stationary points and extrema

10 single-choice questions · 20 minutes · 3 Foundation + 3 Core + 3 Application + 1 optional Challenge. Choose one answer per question.  
Work at your own pace; partial completion is welcome.

### Q1 · Rank 1 / Foundation

Find and classify the stationary point of  $y = -x^2 + x + 1$ .

☐ A Maximum at  $(1/2, 5/4)$ .

☐ B Minimum at  $(1/2, 5/4)$ .

☐ C Maximum at  $(1, 1)$ .

☐ D Minimum at  $(-1/2, 1/4)$ .

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### Q2 · Rank 1 / Foundation

Find and classify the stationary point of  $y = x^2 - 4x + 4$ .

☐ A Maximum at  $(2, 0)$ .

☐ B Minimum at  $(4, 4)$ .

☐ C Minimum at  $(2, 0)$ .

☐ D Minimum at  $(0, 4)$ .

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### Q3 · Rank 1 / Foundation

Find and classify the stationary point of  $y = x^2 - 20x + 105$ .

☐ **A** Minimum at (20, 105).

☐ **B** Maximum at (10, 5).

☐ **C** Minimum at (10, 105).

☐ **D** Minimum at (10, 5).

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### Q4 · Rank 2 / Core

Find and classify the stationary points of  $y = -x^3 + 3x$ .

☐ **A** Minimum  $(-1, -2)$ ; maximum  $(1, 2)$ .

☐ **B** Maximum  $(-1, -2)$ ; minimum  $(1, 2)$ .

☐ **C** Minimum  $(0, 0)$ ; maximum  $(1, 2)$ .

☐ **D** Minimum  $(-1, 2)$ ; maximum  $(1, -2)$ .

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### Q5 · Rank 2 / Core

The demand function is  $P = 40 - 2Q$ . Find the value of  $Q$  which maximises total revenue.

☐ A 20☐ B 10☐ C 40☐ D 0

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### Q6 · Rank 2 / Core

The average cost function is  $AC = 13/Q + Q + 2$ . Find the value of  $Q$  which minimises average cost, for  $Q > 0$ .

☐ A 13☐ B  $2\sqrt{13}$ ☐ C  $\sqrt{13}$ ☐ D  $-\sqrt{13}$ 

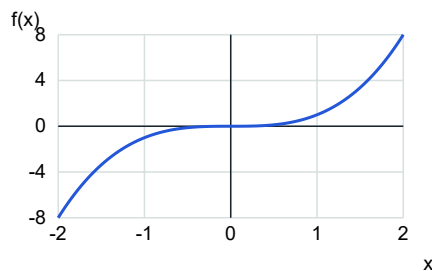
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### Q7 · Rank 3 / Application

For  $f(x) = x^3$ , verify that  $f'(0) = f''(0) = 0$  and classify the stationary point at 0.



☐ A Local maximum.

☐ B Local minimum.

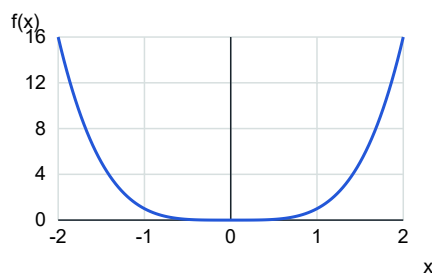
☐ C Both a maximum and a minimum.

☐ D Stationary point of inflection.

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### Q8 · Rank 3 / Application

For  $f(x) = x^4$ , verify that  $f'(0) = f''(0) = 0$  and classify the stationary point at 0.



☐ A Local minimum.

☐ B Local maximum.

☐ C Stationary point of inflection.

☐ D No stationary point.

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### Q9 · Rank 3 / Application

For  $f(x) = -x^6$ , verify that  $f'(0) = f''(0) = 0$  and classify the stationary point at 0.

☐ A Local minimum.

☐ B Local maximum.

☐ C Stationary point of inflection.

☐ D No stationary point.

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### Q10 · Rank 4 / Challenge · OPTIONAL

Daily sales are  $S = t^3 - 24t^2 + 180t + 60$  for  $0 \leq t \leq 13$ . Find the maximum and minimum daily sales during the period between  $t = 5$  and  $t = 9$ . Choose (maximum, minimum).

☐ A (492, 460)

☐ B (485, 465)

☐ C (492, 465)

☐ D (465, 492)

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